INCH-POUND
MIL-DTL-24640/26A
8 November 2011
SUPERSEDING
MIL-DTL-24640/26
9 August 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 TO +105 °C, 600 VOLTS, TYPE 5XO

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight

First	-	Tin-coated copper conductor, AWG size (see table I).
Second	-	Component wires shall be SAE-AS81044/12, except identification shall be in accordance with table II. Testing shall be in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing procedures shall be in accordance with SAE-AS81044.
Third	-	Five components shall be cabled together. Cabling sequence shall be consecutive. Fillers may be used as necessary to form a round cable.
Fourth	-	Optional binder tape.
Fifth	-	Braided shield of 34 AWG tin-coated copper. Braid angle of 40 to 50 degrees; minimum coverage of 90 percent.
Sixth	-	Cross-linked polyolefin jacket with a minimum wall thickness of 0.035 inch. Color black. NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

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TABLE I. Construction details.

Military part no. M24640/26	Type & size	Conductor size AWG	Conductor resistance	Component wire	Minimum bend radius (inch)	Overall diameter (inches)	
						Type 5XO	
						(min.)	(max.)
-01UO	5XO-3	16	4.98	SAE-AS81044/12	2.0	0.300	0.330
-02UO	5XO-4	14	3.18	SAE-AS81044/12	2.5	0.340	0.370
-03UO	5XO-6	12	2.02	SAE-AS81044/12	3.0	0.375	0.410

TABLE II. Color code.

Component	Insulation Color	Print
1	White	Black
2	White	Red
3	White	Orange
4	White	White
5	White	Green

REQUIREMENTS:

Qualification required.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Accelerated aging	Oven temperature 300±2 °C (572±3.6 °F)for 6 hours
Blocking	150±2 °C (302±3.6 °F)
Flammability	30 seconds (max.)
	3 inches (max.)
	No flaming tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test (100% test)	
Primary insulation	Impulse test - 6.0 kV (peak) or 3 kHz spark test - 1.5kV root mean square
Component wire	Impulse test - 8.0 kV (peak) or 3 kHz spark test - 5.5kV root mean square
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	Oven temperature 200±2 °C for 168 hours
Physical properties of primary insulation:	
Tensile strength	2500 pounds per inch (min.)

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Elongation	150 percent (min.)
Shrinkage	0.125 inch (max.) at 300±2 °C (572±3.6 °F)
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	2500
Conductor to shield	2500
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance:	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	Not applicable
Capacitance unbalance (percent, max.)	Not applicable
Characteristic impedance (at 1 MHz, ohms)	Not applicable
Watertightness (see table IX of MIL-DTL-24640 for limits of water leakage)	Not applicable
Weight	No failure

Group B:

Cold bending, cable mandrel diameter 12 times cable diameter	No damage
Drip	Not applicable
Cross-linked proof test (percent, max.)	
Jacket	50
Tear (lb/in thickness, min.)	35

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Physicals (unaged)	
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	Not applicable

Group C:

Physicals (aged), air oven	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Heat distortion (percent of unaged, max.)	30
Permanence of printing (conductor insulation, cycles, Method 1 only, min.)	75
Permanence of printing (jacket) (cycles, min.)	125
Cable sealant removability	Not applicable
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
EMP response (dB, min.)	Not applicable

Group D:

Flame propagation (cable)	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2

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Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	35
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
5XO-3, 5XO-4, 5XO-6	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-101)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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